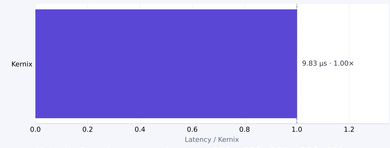
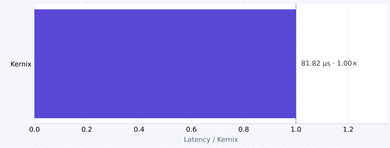
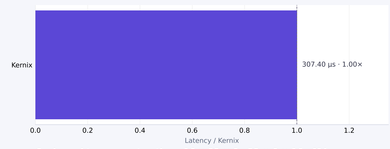
### Matmul · 2 workers · reused output · 64 × 64 · 64 × 64 · f64



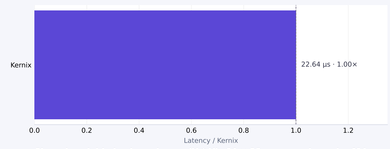

### Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f64



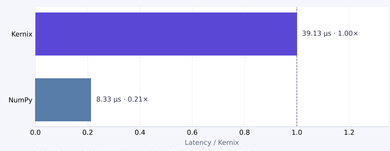

### Matmul · 2 workers · reused output · 256 × 256 · 256 × 256 · f64



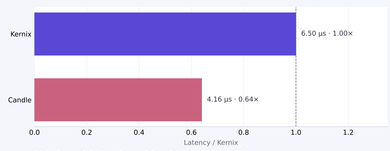

### Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f32



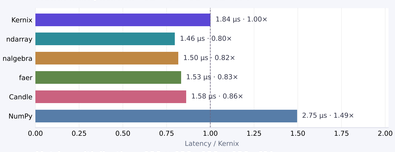

### Coalesced tensor contraction · 16 × 32 × 8 · 32 × 8 × 16 · f64



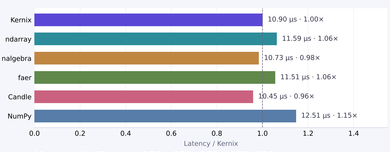

### Shared-variable backward · sum(X * X), 64 × 32; prepared graph · f64



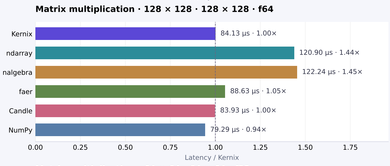

### Matrix multiplication · 32 × 32 · 32 × 32 · f64



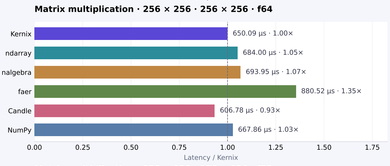

### Matrix multiplication · 64 × 64 · 64 × 64 · f64



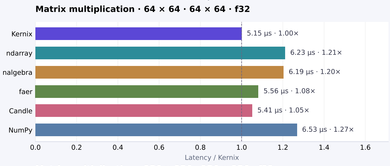

### Matrix multiplication · 128 × 128 · 128 × 128 · f64



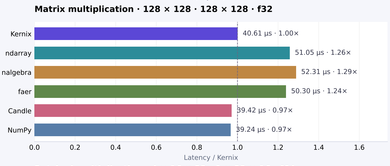

### Matrix multiplication · 256 × 256 · 256 × 256 · f64



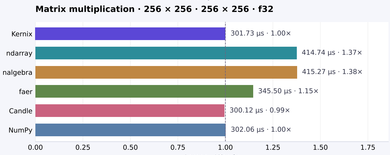

### Matrix multiplication · 64 × 64 · 64 × 64 · f32



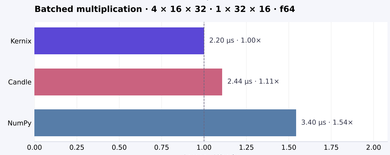

### Matrix multiplication · 128 × 128 · 128 × 128 · f32



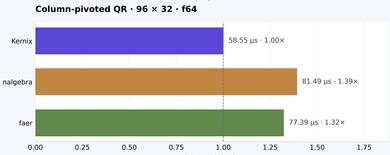

### Matrix multiplication · 256 × 256 · 256 × 256 · f32



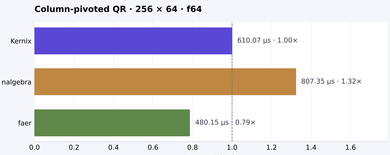

### Batched multiplication · 4 × 16 × 32 · 1 × 32 × 16 · f64



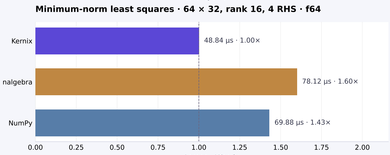

### Column-pivoted QR · 96 × 32 · f64



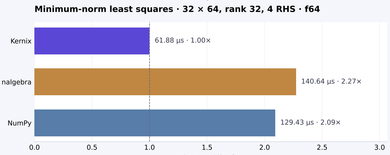

### Column-pivoted QR · 256 × 64 · f64



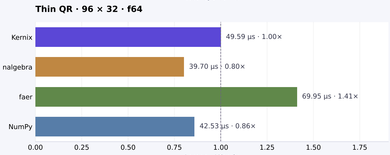

### Minimum-norm least squares · 64 × 32, rank 16, 4 RHS · f64



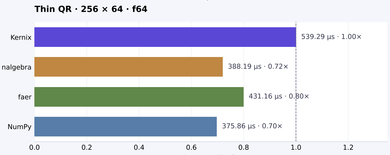

### Minimum-norm least squares · 32 × 64, rank 32, 4 RHS · f64



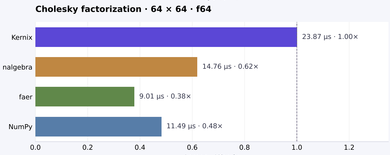

### Thin QR · 96 × 32 · f64



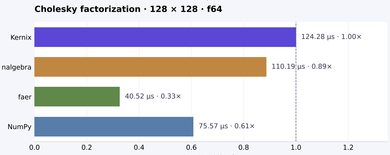

### Thin QR · 256 × 64 · f64



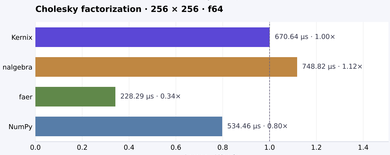

### Cholesky factorization · 64 × 64 · f64



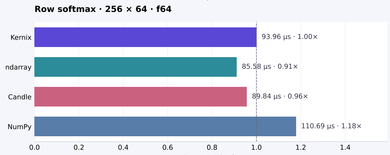

### Cholesky factorization · 128 × 128 · f64



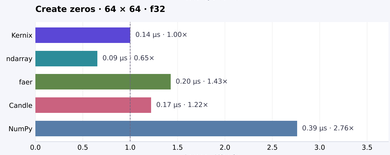

### Cholesky factorization · 256 × 256 · f64



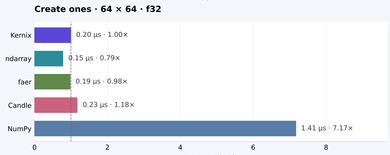

### Row softmax · 256 × 64 · f64



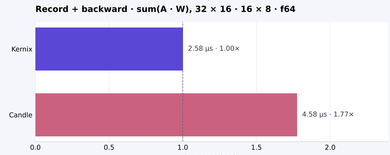

### Create zeros · 64 × 64 · f32



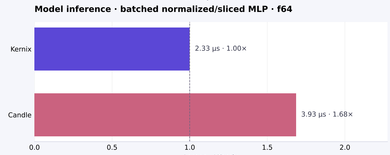

### Create ones · 64 × 64 · f32



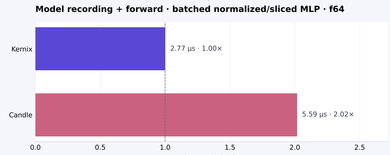

### Record + backward · sum(A · W), 32 × 16 · 16 × 8 · f64



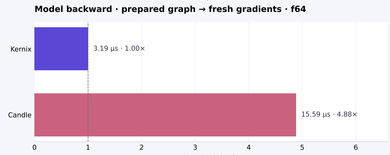

### Model inference · batched normalized/sliced MLP · f64




### Model recording + forward · batched normalized/sliced MLP · f64




### Model backward · prepared graph → fresh gradients · f64




### Parameter update · transactional SGD, fixed gradients · f64